DOCUMENT 05

Feature Specification

Feature-by-feature objectives, flows, requirements, and acceptance criteria

Study Topic Extractor

Version: 1.0
Status: Initial Specification
Date: September 28, 2026

Turn long study material into a focused study roadmap.

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1. Feature Catalog

Twenty-two features, each with objective, value, flow, inputs, outputs, backend/frontend/database requirements, edge cases, error states and acceptance criteria. Priority levels and thresholds are defined once in the NLP/AI Specification and are not restated with different values here.

ID	Feature	Phase
F01	Text Input	MVP
F02	PDF Upload	MVP
F03	PDF Text Extraction	MVP
F04	OCR	Phase 2
F05	Text Preprocessing	MVP
F06	Keyword Extraction	MVP
F07	N-gram Extraction	MVP
F08	TF-IDF Analysis	MVP
F09	Candidate Topic Generation	MVP
F10	Topic Ranking	MVP
F11	Priority Classification	MVP
F12	Semantic Similarity	Phase 2
F13	Topic Clustering	Phase 2
F14	Topic Hierarchy	Phase 2
F15	Topic Relationship Graph	Phase 2
F16	Topic Summaries	MVP
F17	Exam Focus Mode	Phase 2
F18	Question Generation	Phase 2
F19	Export	MVP
F20	Document History	Phase 2
F21	User Dashboard	Phase 2
F22	Settings	MVP (local) / Phase 2 (account)

1.1 MVP Feature Set

F01, F02, F03, F05, F06, F07, F08, F09, F10, F11, F16, F19 and Settings (local) form the MVP. Everything else is Phase 2 and must not be started before MVP acceptance criteria pass.

2. Feature Traceability

Feature	PRD requirement	Endpoint	Test class (see Testing doc)
F01 Text Input	FR-01	POST /api/analyze	UT-INPUT, E2E-01
F02, F03 PDF	FR-02	POST /api/upload	IT-PDF, E2E-01
F05 - F11	FR-03 to FR-07	POST /api/analyze	UT-NLP, IT-PIPE
F16 Summaries	FR-08	POST /api/analyze	UT-SUM
F19 Export	FR-10	POST /api/export	IT-EXPORT, E2E-03
F12 - F15	FR-11, FR-12	POST /api/analyze (P2 flags)	UT-SEM, IT-GRAPH
F17, F18	FR-13	POST /api/documents/{id}/questions	IT-Q, E2E-04
F04 OCR	FR-14	POST /api/upload	IT-OCR
F20 - F22	FR-15	documents, settings	IT-DB, E2E-05

3. Text Input

Feature ID: F01 **Phase:** MVP

Aspect	Specification
Objective	Let users paste or type study material for analysis.
User value	Zero-friction start without needing a file.
User flow	Open Analyze > Paste Text tab > enter text (optional title) > Extract Topics.
Inputs	text (200 to 150,000 chars), title (optional, max 120 chars)
Outputs	Validated payload sent to analysis.
Backend requirements	`POST /api/analyze` validates length, strips control characters, normalizes Unicode (NFKC).
Frontend requirements	Textarea with live counter, clear button, sample-text button, disabled state while busy.
Database requirements	None in MVP. P2: `documents` row (source_type=text) when save=true.
Edge cases	Only whitespace; emoji/RTL text; pasted HTML (tags stripped); 1 MB clipboard paste.
Error states	TEXT_TOO_SHORT, DOCUMENT_TOO_LARGE, INVALID_REQUEST.
Acceptance criteria	Button enabled only within bounds; counter accurate; sample text produces results in < 3 s.

4. PDF Upload

Feature ID: F02 **Phase:** MVP

Aspect	Specification
Objective	Accept PDF files safely.
User value	Analyze lecture PDFs without copying text.

Aspect	Specification
User flow	Analyze > Upload PDF > drop/select file > validation > extraction preview.
Inputs	file (application/pdf, <= 10 MB)
Outputs	Validated file bytes in memory.
Backend requirements	`POST /api/upload`: extension, MIME and `%PDF-` magic-byte checks; size limit; streamed read with cap.
Frontend requirements	Drop zone, file chip, progress bar, remove button, error alerts.
Database requirements	None in MVP. P2: Storage object + `documents` row.
Edge cases	Renamed non-PDF; 0-byte file; double extension; huge file; duplicate upload.
Error states	INVALID_FILE_TYPE, INVALID_PDF, FILE_TOO_LARGE, RATE_LIMITED.
Acceptance criteria	Non-PDFs rejected client and server side; valid PDF proceeds; no file written to disk in MVP.

5. PDF Text Extraction

Feature ID: F03 **Phase:** MVP

Aspect	Specification
Objective	Convert PDF pages into clean text with page mapping.
User value	Reliable input for NLP.
User flow	Automatic after upload validation.
Inputs	PDF bytes
Outputs	text, page_count, quality (0-1), warnings, page offsets.
Backend requirements	`pdf_processor.py` with PyMuPDF `page.get_text("text", sort=True)`; hyphenation repair; header/footer removal; page limit 200.
Frontend requirements	Preview panel (first 2,000 chars) and quality alert.
Database requirements	P2: `documents.extracted_text`, `page_count`, `extraction_quality`, `extraction_method=native`.
Edge cases	Multi-column layouts; tables; footnotes; ligatures; encrypted; corrupted; tiny text layer.
Error states	PASSWORD_PROTECTED_PDF, NO_READABLE_TEXT, DOCUMENT_TOO_LARGE, INVALID_PDF.
Acceptance criteria	Text-based 50-page PDF extracts in < 5 s; quality reported; encrypted PDFs give correct error.

6. OCR

Feature ID: F04 **Phase:** Phase 2

Aspect	Specification
Objective	Extract text from scanned pages.
User value	Support photographed or scanned notes.
User flow	Automatic when quality < 0.2 and `ENABLE_OCR=true`, or user chooses Try OCR.

Aspect	Specification
Inputs	Pages without text layer
Outputs	Recognized text, `extraction_method=ocr`.
Backend requirements	`ocr_service.py`: render pages at 200 DPI via PyMuPDF, run Tesseract (English), cap 50 pages, per-page timeout 15 s.
Frontend requirements	Option banner: Scanned document detected. Try OCR (takes longer).
Database requirements	`documents.extraction_method='ocr'`.
Edge cases	Handwriting; skewed pages; low resolution; mixed native + scanned.
Error states	OCR_UNAVAILABLE (503), NO_READABLE_TEXT.
Acceptance criteria	A clean scanned page yields >= 90% character accuracy; OCR never runs unless enabled.

7. Text Preprocessing

Feature ID: F05 **Phase:** MVP

Aspect	Specification
Objective	Normalize text and produce tokens, lemmas, sentences.
User value	Consistent analysis and deduplicated counting.
User flow	Internal pipeline stage.
Inputs	Raw text
Outputs	Sentences, tokens with lemma and POS, cleaned text, character offsets.
Backend requirements	`text_cleaner.py` and `text_preprocessor.py` (spaCy).
Frontend requirements	None.
Database requirements	P2: `document_sections` store section boundaries.
Edge cases	Bullets; numbered lists; code snippets; URLs; tables flattened; Unicode dashes.
Error states	PROCESSING_FAILED on unexpected error.
Acceptance criteria	Deterministic output; sentence boundaries correct on 95% of gold sentences; offsets map back to original text.

8. Keyword Extraction

Feature ID: F06 **Phase:** MVP

Aspect	Specification
Objective	Extract important terms and phrases.
User value	Quick vocabulary scan.
User flow	Internal stage; results in Keywords tab and chips.
Inputs	Candidates + TF-IDF

Aspect	Specification
Outputs	Top 30 keywords with score and count.
Backend requirements	`keyword_extractor.py`: rank candidates by TF-IDF x length boost; filter stop-heavy phrases.
Frontend requirements	Keyword chips with counts; click to filter topics containing the keyword.
Database requirements	P2: `keywords` table.
Edge cases	Single-word generic terms; acronyms; hyphenated terms.
Error states	-
Acceptance criteria	Example: `Supervised learning uses labeled datasets to train machine learning models` yields Supervised Learning, Labeled Dataset, Machine Learning.

9. N-gram Extraction

Feature ID: F07 **Phase:** MVP

Aspect	Specification
Objective	Generate unigram, bigram and trigram candidates and prefer meaningful phrases.
User value	Captures concepts like support vector machine.
User flow	Internal stage.
Inputs	Lemmatized tokens with POS
Outputs	Candidate list with counts, positions, n-gram size.
Backend requirements	`ngram_extractor.py`: POS pattern (ADJ|NOUN)* NOUN; edge stopword filter; min count 2 for n >= 2 unless in heading/definition; NPMI >= 0.1; subsumption rule.
Frontend requirements	None.
Database requirements	None.
Edge cases	Overlapping phrases; acronyms (SVM); capitalized proper nouns; hyphenation.
Error states	-
Acceptance criteria	support vector machine appears as one trigram topic; machine and learning alone are suppressed when they occur almost always inside machine learning.

10. TF-IDF Analysis

Feature ID: F08 **Phase:** MVP

Aspect	Specification
Objective	Measure term distinctiveness.
User value	Highlights what is specific to this document.
User flow	Internal stage.
Inputs	Candidates, sections

Aspect	Specification
Outputs	TF-IDF per candidate normalized 0-1.
Backend requirements	`tfidf_service.py` with scikit-learn `TfidfVectorizer` (custom analyzer) blended with `data/background_idf.json`.
Frontend requirements	Score breakdown bar in topic detail.
Database requirements	P2: stored in `topics.components`.
Edge cases	Single-section documents; very small documents; identical sections.
Error states	-
Acceptance criteria	Same input yields same TF-IDF; background IDF used when sections < 5.

11. Candidate Topic Generation

Feature ID: F09 **Phase:** MVP

Aspect	Specification
Objective	Turn candidates into topic candidates.
User value	Ensures topics are meaningful study concepts, not fragments.
User flow	Internal stage.
Inputs	Keywords, n-grams, headings, definition sentences
Outputs	Deduplicated topic candidates (max 60).
Backend requirements	`topic_extractor.py`: merge by normalized lemma key, add heading terms, definition subjects, capitalized terms; drop generic words.
Frontend requirements	None.
Database requirements	None.
Edge cases	Synonyms (P2 embeddings); plural/singular variants; abbreviations.
Error states	-
Acceptance criteria	Expected topics from the spec example are all present; no stopword-only candidates.

12. Topic Ranking

Feature ID: F10 **Phase:** MVP

Aspect	Specification
Objective	Score every topic 0-100.
User value	Tells the student what matters most.
User flow	Internal stage; visible as rank and score.
Inputs	Candidates with component signals
Outputs	Ordered topics with score and component values.

Aspect	Specification
Backend requirements	`topic_ranker.py`: weighted sum of TF-IDF, frequency, position, semantic (weights renormalized over enabled signals).
Frontend requirements	Sorted list, score bar, breakdown in drawer, rank number.
Database requirements	P2: `topics.score`, `rank`, `components`.
Edge cases	Ties (break by occurrences then alphabetical); all components equal; one candidate.
Error states	-
Acceptance criteria	Worked example (0.82, 0.90, 0.75, 0.88 with full weights) yields 83.5; identical input yields identical order.

13. Priority Classification

Feature ID: F11 **Phase:** MVP

Aspect	Specification
Objective	Map score to HIGH / MEDIUM / LOW.
User value	Clear study order.
User flow	Automatic.
Inputs	Score, thresholds
Outputs	priority label per topic.
Backend requirements	`priority_classifier.py`: HIGH >= 80, MEDIUM 50-79, LOW < 50 (configurable).
Frontend requirements	Priority badge with icon + label; filter chips with counts.
Database requirements	P2: `topics.priority`.
Edge cases	Score exactly 80 or 50; thresholds changed by user; no HIGH topics.
Error states	INVALID_REQUEST if thresholds invalid (high <= medium).
Acceptance criteria	Boundary tests pass; label never shown without icon and text.

14. Semantic Similarity

Feature ID: F12 **Phase:** Phase 2

Aspect	Specification
Objective	Understand meaning beyond exact words.
User value	Better relevance and relations.
User flow	Internal stage.
Inputs	Topic strings, sections
Outputs	Embeddings, cosine similarity matrix, semantic relevance signal.

Aspect	Specification
Backend requirements	`semantic_analyzer.py` with `all-MiniLM-L6-v2`; normalize vectors; relevance = cosine to document centroid.
Frontend requirements	None directly; semantic component in breakdown.
Database requirements	P2: `topics.components.semantic`; relationships table.
Edge cases	Model load failure; very short topic strings; memory pressure.
Error states	SERVICE_UNAVAILABLE (fallback: disable semantic and add warning `SEMANTIC_DISABLED`).
Acceptance criteria	Predicting categories is closer to Classification than to Clustering (cosine ordering test).

15. Topic Clustering

Feature ID: F13 **Phase:** Phase 2

Aspect	Specification
Objective	Group related topics.
User value	Reveals subject structure.
User flow	Internal stage.
Inputs	Topic embeddings
Outputs	Clusters with labels and cohesion.
Backend requirements	`topic_clusterer.py`: agglomerative (average linkage, cosine, threshold 0.45); K-Means fallback k = clamp(round(sqrt(n/2)), 2, 8).
Frontend requirements	Cluster chips and grouping option in results.
Database requirements	`topic_clusters`; `topics.cluster_id`.
Edge cases	Fewer than 6 topics (skip); one giant cluster; singleton clusters.
Error states	-
Acceptance criteria	Machine Learning terms cluster apart from unrelated terms in the evaluation document.

16. Topic Hierarchy

Feature ID: F14 **Phase:** Phase 2

Aspect	Specification
Objective	Create parent-child structure.
User value	Shows what belongs under what.
User flow	Internal stage; Hierarchy tab.
Inputs	Ranked topics, sentences, embeddings
Outputs	Tree (max depth 4) with `parent_id`, `children_ids`.

Aspect	Specification
Backend requirements	`hierarchy_builder.py`: lexical head match, pattern rules (types of, such as, includes), semantic subsumption; NetworkX maximum spanning arborescence; no cycles.
Frontend requirements	Collapsible tree with priority badges.
Database requirements	`topics.parent_topic_id`, `depth`; hierarchy rows in `topic_relationships`.
Edge cases	Multiple plausible parents; cycles; orphan topics (top-level); depth overflow.
Error states	-
Acceptance criteria	Every topic has 0 or 1 parent; no cycles; Supervised Learning is a child of Machine Learning on the sample.

17. Topic Relationship Graph

Feature ID: F15 **Phase:** Phase 2

Aspect	Specification
Objective	Visualize connections.
User value	Aids conceptual understanding.
User flow	Results > Graph tab.
Inputs	Topics, relations
Outputs	Graph JSON: nodes (id, label, score, priority, cluster) and edges (source, target, type, weight).
Backend requirements	`graph_builder.py` (NetworkX): hierarchy edges + semantic edges (cosine >= 0.55, top 3 per node); cap 40 nodes.
Frontend requirements	Cytoscape canvas, legend, edge toggles, list alternative.
Database requirements	`topic_relationships`.
Edge cases	Disconnected components; hub nodes; many nodes.
Error states	-
Acceptance criteria	Graph renders < 1 s for 40 nodes; keyboard focus and list alternative exist.

18. Topic Summaries

Feature ID: F16 **Phase:** MVP

Aspect	Specification
Objective	Give a short explanation per topic.
User value	Understand a topic without rereading.
User flow	Automatic; shown on topic card.
Inputs	Topic, sentences
Outputs	1 to 2 sentence summary (<= 280 chars) and optional definition.

Aspect	Specification
Backend requirements	`summarizer.py`: extractive sentence scoring; optional LLM hybrid (P2) with strict grounding; fallback always extractive.
Frontend requirements	Summary in card and drawer; copy button.
Database requirements	`topics.summary`, `definition`.
Edge cases	No sentence contains the topic; only fragments; duplicate sentences.
Error states	-
Acceptance criteria	100% of returned topics have a summary that is a substring (extractive) of the source; summary <= 280 chars.

19. Exam Focus Mode

Feature ID: F17 **Phase:** Phase 2

Aspect	Specification
Objective	Revision-oriented views.
User value	Efficient exam preparation.
User flow	Results > Exam Focus > pick mode.
Inputs	Analysis Result, mode
Outputs	Mode-specific content.
Backend requirements	No new pipeline; uses stored result; question endpoint for Deep Study.
Frontend requirements	Segmented control, checklists, print styles.
Database requirements	`study_sessions`.
Edge cases	No HIGH topics; missing hierarchy.
Error states	NOT_FOUND, PROCESSING_FAILED.
Acceptance criteria	Quick, Normal and Deep show exactly the content defined in the UI/UX specification.

20. Question Generation

Feature ID: F18 **Phase:** Phase 2

Aspect	Specification
Objective	Produce possible exam questions.
User value	Self-testing.
User flow	Deep Study > Generate questions.
Inputs	Topics, hierarchy, sentences
Outputs	3 to 6 questions per HIGH topic with type and difficulty.

Aspect	Specification
Backend requirements	`question_generator.py`: templates by topic role; sibling comparison; optional LLM paraphrase.
Frontend requirements	Question list with type badges.
Database requirements	`generated_questions`.
Edge cases	Sparse context; topics with no siblings.
Error states	LLM_UNAVAILABLE (fallback to templates silently with warning).
Acceptance criteria	Every question references a topic in the result; no duplicates; works with LLM disabled.

21. Export

Feature ID: F19 **Phase:** MVP

Aspect	Specification
Objective	Download results.
User value	Offline revision and sharing.
User flow	Results > Export > format > download.
Inputs	Analysis Result, format, options
Outputs	File download.
Backend requirements	`POST /api/export` (MVP, stateless); `exporter.py`; CSV injection guard.
Frontend requirements	Export modal, toast feedback.
Database requirements	None (P2: `/api/documents/{id}/export`).
Edge cases	Empty topics; long summaries; special characters.
Error states	INVALID_REQUEST (unknown format), PROCESSING_FAILED.
Acceptance criteria	PDF, CSV, TXT open correctly; file name `study-roadmap-{slug}-{date}.{ext}`.

22. Document History

Feature ID: F20 **Phase:** Phase 2

Aspect	Specification
Objective	Revisit saved analyses.
User value	Continuity across sessions.
User flow	History page.
Inputs	Authenticated user
Outputs	Paginated list.
Backend requirements	`GET /api/documents` with pagination (page size 20), search by title.

Aspect	Specification
Frontend requirements	Table/cards, search, sort, delete.
Database requirements	`documents`, `analysis_runs`.
Edge cases	No documents; deletion of the currently open document.
Error states	UNAUTHORIZED, NOT_FOUND.
Acceptance criteria	Only the owner's documents are listed; delete cascades fully.

23. User Dashboard

Feature ID: F21 **Phase:** Phase 2

Aspect	Specification
Objective	Personal overview.
User value	Motivation and quick access.
User flow	Dashboard page.
Inputs	Authenticated user
Outputs	Recent documents, study stats.
Backend requirements	Aggregated queries over `documents`, `study_sessions`.
Frontend requirements	Stat cards, lists.
Database requirements	Reads only.
Edge cases	New user with no data.
Error states	UNAUTHORIZED.
Acceptance criteria	Empty state for new users; stats match session records.

24. Settings

Feature ID: F22 **Phase:** MVP (local) / Phase 2 (account)

Aspect	Specification
Objective	Configure preferences.
User value	Personalization.
User flow	Settings page.
Inputs	Theme, default mode, max topics, weights, thresholds
Outputs	Saved preferences applied to analyze options.
Backend requirements	P2: `GET/PATCH /api/settings`. Weights must sum to 1.0 (+/- 0.001); thresholds high > medium.
Frontend requirements	Form with live validation; reset to defaults.

Aspect	Specification
Database requirements	MVP: localStorage. P2: `users.settings` JSONB.
Edge cases	Invalid weights; missing storage.
Error states	INVALID_REQUEST.
Acceptance criteria	Changing weights changes ordering deterministically; defaults restorable.